



# GenomelQ — Per-Page Guide Index

This folder contains a deep-dive guide for **every page** in the GenomelQ web app. Each guide follows the same structure:

- What it is
- What it tells you
- How it works under the hood
- How to use it
- Practical tips
- End-to-end flow diagram
- Cheat sheet
- Related documents



## Available Guides

| # | Guide                               | Page                                                                                         | Format   |
|---|-------------------------------------|----------------------------------------------------------------------------------------------|----------|
| 1 | <a href="#">Analyze Page</a>        | <b>Analyze</b> — DNA prediction with verdict, saliency, OOD, similarity, ORFs, circular plot | PDF      |
| 2 | <a href="#">Copilot Page</a>        | <b>Copilot</b> — RAG chatbot with Gemini / Ollama / Demo providers                           | Markdown |
| 3 | <a href="#">Mutation Lab Page</a>   | <b>Mutation Lab</b> — reference vs variant comparison with AA effect classification          | Markdown |
| 4 | <a href="#">Batch Upload Page</a>   | <b>Batch Upload</b> — multi-record FASTA classification                                      | Markdown |
| 5 | <a href="#">Knowledge Base Page</a> | <b>Knowledge Base</b> — disease and gene reference cards                                     | Markdown |
| 6 | <a href="#">User Manual Page</a>    | <b>User Manual</b> — in-app help (the meta-guide)                                            | Markdown |
| 7 | <a href="#">Presentation Page</a>   | <b>Presentation</b> — 12-slide demo deck                                                     | Markdown |
| 8 | <a href="#">About Page</a>          | <b>About</b> — project description + author + disclaimer                                     | Markdown |

Plus the project-wide test plan: | [Testing Guide](#) | All pages — full manual test plan with ~50 test cases | PDF |



## How to use these guides

For **first-time users** of GenomelQ

Start with the [Analyze Page](#) guide. It walks you through the most-used page step-by-step.

For **defending your project**

Read the [Presentation Page](#) guide for the suggested 12-slide narration script.

For **writing your thesis or report**

- [Analyze Page](#) for the methodology deep-dive
- [Copilot Page](#) for the RAG architecture explanation
- [Mutation Lab Page](#) for the bioinformatics layer

For **debugging or extending the project**

Each guide's  *How it works under the hood* section names the exact source files and functions, so you can navigate from a feature to its code.

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 **Convert Markdown to PDF**

To match the existing `ANALYZE_PAGE_GUIDE.pdf` and `TESTING_GUIDE.pdf` style:

Option 1 — VS Code extension (recommended)

1. Install the **Markdown PDF** extension by yzane.
2. Open any `.md` file.
3. **Right-click** → **Markdown PDF: Export (pdf)**.
4. The PDF is saved next to the source file.

Option 2 — Pandoc

```
pandoc COPILOT_PAGE_GUIDE.md -o COPILOT_PAGE_GUIDE.pdf
```

Option 3 — Browser print

1. Open the `.md` in a Markdown viewer (e.g., `marked` extension).
2. Print to PDF (`Ctrl+P` → Save as PDF).

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 **Project-wide Documents (in the project root)**

| Document                                 | Purpose                                              |
|------------------------------------------|------------------------------------------------------|
| <code>README.md</code>                   | Public-facing project overview (GitHub README)       |
| <code>DOCS.md</code>                     | Comprehensive technical reference (12 sections)      |
| <code>HOW_TO_RUN_IN_VSCODE.md</code>     | Step-by-step VS Code setup                           |
| <code>TESTING_GUIDE.md</code>            | Full manual test plan                                |
| <code>figures/SCREENSHOT_GUIDE.md</code> | Instructions for capturing the 12 README screenshots |

*Last updated as part of the GenomeIQ Step 7 documentation pass.*